

React vs Next.js Cheat Sheet

This cheat sheet highlights the key differences and additions in Next.js compared to plain React.

Routing

Feature	React	Next.js
Routing setup	Manual with react-router-dom	File-based (auto) routing
Example route	<code><Route path="/about" element={<About />} /></code>	<code>pages/about.js</code> auto becomes <code>/about</code>
Link navigation	<code><Link to="/about"></code>	<code><Link href="/about"></code> from <code>next/link</code>

Image Handling

Feature	React	Next.js
Image import	<code></code>	<code><Image src="..." /></code> from <code>next/image</code>
Optimization	Manual	Built-in optimization (lazy loading, resizing)

API Handling

Feature	React	Next.js
Backend/API code	Separate Express/Node server needed	Use <code>pages/api/hello.js</code> for serverless API routes
Example	❌ (not built-in)	✅ <code>export default function handler(req, res) {...}</code>

Rendering Modes

Rendering	React	Next.js
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CSR (Client-Side Rendering)	✓ Default	✓
SSR (Server-Side Rendering)	Manual setup	✓ <code>getServerSideProps()</code>
SSG (Static Site Generation)	Manual with build tools	✓ <code>getStaticProps()</code> / <code>getStaticPaths()</code>
ISR (Incremental Static Regeneration)	✗	✓ Revalidate static pages

Routing Hooks

Feature	React	Next.js
Navigation	<code>useNavigate()</code>	<code>useRouter()</code>
Example	<code>useNavigate('/home')</code>	<code>router.push('/home')</code>

Head Management

Feature	React	Next.js
Page title/meta	<code><Helmet></code> from <code>react-helmet</code>	<code><Head></code> from <code>next/head</code>

Folder Structure

React

```

/src
  /components
  /pages (optional)
App.js
index.js

```

Next.js

```

/pages
  index.js → Homepage
  about.js → /about
  /api/hello.js → API Route
/public

```

/components

Bonus: Next.js Commands

Task	Command
Start dev server	npm run dev
Build app	npm run build
Start production server	npm start
Export static site	next export